

KEESLER AFB, MS, USA

WMO: 747686

Lat: 30.417N

Lon: 88.917W

Elev: 33

StdP: 14.68

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13820

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.2	34.2	14.1	11.3	35.6	19.4	14.6	40.9	21.0	48.4	19.7	48.1	9.0	20	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	94.3	80.0	92.4	79.5	90.8	79.0	83.5	90.2	82.1	88.5	81.1	87.4	9.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.6	164.5	87.2	80.8	160.3	86.3	79.2	151.8	84.8	47.2	90.1	45.7	88.8	44.5	87.4	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.8	16.8	15.4	DB	22.8	97.6	6.4	2.9	18.2	99.7	14.4	101.5	10.8	103.1	6.1	105.2
			WB	20.2	84.1	5.9	1.9	16.0	85.5	12.6	86.6	9.3	87.7	5.0	89.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.1	51.4	56.1	61.5	68.5	76.0	81.6	83.1	83.2	79.5	70.7	60.8	55.6
	DBStd	13.00	9.34	8.97	7.99	6.15	4.85	3.29	2.87	3.15	4.10	7.03	7.86	8.86
	HDD50	205	98	41	11	0	0	0	0	0	0	10	45	
	HDD65	1402	427	265	161	37	2	0	0	0	0	31	169	310
	CDD50	7166	140	213	367	554	807	949	1027	1028	886	640	335	220
	CDD65	2890	5	17	52	141	344	499	562	563	436	207	45	19
	CDH74	28510	2	15	114	625	2988	5656	6559	6540	4321	1524	148	18
	CDH80	10814	0	0	4	45	784	2292	2867	2874	1605	335	8	0
(14) Wind	WSAvg	7.4	8.0	8.2	8.4	8.4	7.8	7.2	6.2	6.1	6.8	7.2	7.0	7.6
(15) Precipitation	PrecAvg	65.4	5.6	4.2	5.9	4.8	5.2	6.9	7.5	6.4	6.6	3.6	4.4	4.8
	PrecMax	90.5	18.5	8.8	12.2	15.9	12.6	18.1	14.6	13.0	29.6	8.8	13.3	16.3
	PrecMin	42.2	0.0	0.3	0.2	0.6	0.6	1.6	2.4	3.1	1.5	0.0	0.0	1.5
	PrecStd	12.0	4.3	2.5	3.3	3.5	3.4	3.9	3.2	2.3	6.1	2.7	3.2	3.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.5	76.3	79.8	83.9	91.0	94.4	97.1	97.4	94.2	88.6	81.3	76.3
		MCWB	66.1	69.8	69.6	71.1	77.8	78.4	82.5	81.4	77.4	76.6	73.8	70.9
	2%	DB	69.8	72.7	77.1	81.2	88.4	92.0	93.7	94.6	91.1	85.9	77.8	73.3
		MCWB	65.8	67.9	69.1	72.2	76.0	78.9	80.4	80.6	77.5	75.3	70.5	69.6
	5%	DB	67.8	70.3	74.7	79.4	86.2	90.5	91.5	92.3	89.7	83.7	75.3	71.0
		MCWB	65.2	66.3	68.5	71.1	75.0	78.7	79.7	80.2	77.1	73.7	68.2	67.9
	10%	DB	65.5	68.2	72.4	77.6	84.3	88.9	90.2	90.4	87.8	81.5	73.0	68.6
		MCWB	63.0	64.8	67.5	70.1	74.4	78.2	79.2	79.5	76.7	73.3	67.3	66.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.2	72.0	73.4	77.4	81.4	83.5	85.7	85.4	83.4	80.7	76.8	73.6
		MCDB	70.1	74.5	75.8	80.8	87.5	88.4	93.5	92.5	88.6	83.5	78.8	74.4
	2%	WB	67.6	69.9	71.9	75.6	78.7	82.0	83.0	83.7	81.3	78.4	73.1	70.7
		MCDB	68.6	71.7	74.7	78.7	84.3	87.7	90.4	91.0	85.3	82.6	75.9	72.0
	5%	WB	65.9	68.0	70.3	74.3	77.4	80.8	81.5	82.2	79.4	77.0	71.1	68.8
		MCDB	67.2	69.4	73.1	77.4	83.3	87.0	88.3	89.0	85.7	81.8	74.0	70.4
	10%	WB	63.3	66.1	68.9	72.8	76.3	79.8	80.5	81.0	78.4	75.3	69.1	66.9
		MCDB	64.6	67.4	71.5	76.2	82.1	86.3	87.3	87.7	84.7	80.0	71.7	68.4
(35) Mean Daily Temperature Range	5% DB	MDBR	16.5	15.8	15.7	15.0	14.3	13.5	13.6	13.9	15.4	17.5	18.1	15.9
		MCDBR	14.4	13.9	14.4	14.1	15.3	14.7	15.2	15.7	16.9	16.6	17.1	15.2
		MCWBR	11.1	10.2	9.0	8.0	6.9	6.2	6.6	6.5	7.0	8.4	10.7	11.8
	5% WB	MCDBR	12.4	11.5	12.2	11.6	12.7	13.1	14.2	14.4	14.1	13.8	14.1	13.7
		MCWBR	10.8	9.7	8.6	8.2	6.7	6.6	7.0	7.1	7.1	8.2	11.0	11.9
(40) Clear-Sky Solar Irradiance	taub		0.334	0.350	0.374	0.405	0.438	0.465	0.515	0.505	0.453	0.385	0.354	0.342
	taud		2.542	2.502	2.428	2.344	2.286	2.254	2.141	2.166	2.275	2.454	2.485	2.533
	Ebn at Noon		285	290	289	282	272	263	250	251	261	274	275	276
	Edh at Noon		27	31	36	40	43	44	49	47	41	32	29	26
(44) All-Sky Solar Radiation	RadAvg		961	1136	1530	1869	2060	1991	1876	1776	1658	1444	1104	851
	RadStd		80	159	125	152	113	150	132	133	110	151	97	74

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+247	+162

Nomenclature: See separate page